Six Ways Across the Membrane

Draw each one on the membrane strip. Show the particles on both sides, an arrow for the way they move, and the protein or vesicle if one is used. Then tick the two boxes.

Words to use: concentration gradient · high → low · low → high · passive · active · ATP · channel protein · protein pump · aquaporin · vesicle · selectively permeable · equilibrium

1 Simple diffusion

Oxygen slips straight between the phospholipids.

OUTSIDE THE CELL



INSIDE THE CELL

ATP? ☐ yes ☐ no Moves ☐ high → low ☐ low → high

2 Facilitated diffusion

An ion or a sugar goes through a channel or carrier protein.

OUTSIDE THE CELL



INSIDE THE CELL

ATP? ☐ yes ☐ no Moves ☐ high → low ☐ low → high

3 Osmosis

Water moves through an aquaporin.

OUTSIDE THE CELL



INSIDE THE CELL

ATP? ☐ yes ☐ no Moves ☐ high → low ☐ low → high

4 Active transport

A protein pump moves sodium against its gradient.

OUTSIDE THE CELL



INSIDE THE CELL

ATP? ☐ yes ☐ no Moves ☐ high → low ☐ low → high

5 Endocytosis

The membrane folds in around a large particle and pinches off.

OUTSIDE THE CELL



INSIDE THE CELL

ATP? ☐ yes ☐ no Cargo goes ☐ into ☐ out of the cell

6 Exocytosis

A vesicle merges with the membrane and releases insulin outside.

OUTSIDE THE CELL



INSIDE THE CELL

ATP? ☐ yes ☐ no Cargo goes ☐ into ☐ out of the cell

Exit line. The ones that cost the cell ATP are _____ because _____.

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